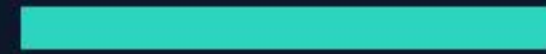


STRATEGIC DATA ANALYSIS

CITY-LEVEL SALES PREDICTION

Using Citizen Characteristics to Forecast Performance and
Identify Regional Segments



PHASE 1: DATASET DESCRIPTION

Overview of data origins and initial feature profiling

| DATASET CHARACTERISTICS

- ✓ **Coverage:** 1,250 observations across 50 metropolitan cities.
- ✓ **Features:** Median Citizen Age, Average Income, and High Education percentages.
- ✓ **Target:** Annual Product Sales Revenue (USD).
- ✓ **Context:** Includes urban infrastructure and housing index metrics.

 Dataset Preview

| ENGINEERING STRATEGY

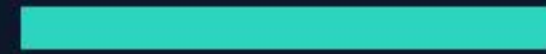
To improve model sensitivity, we derived several composite indices from citizen features:

- ✓ **Wealth Index:** Weighted income and property values.
- ✓ **Tech Affinity:** Correlation between education and high-speed infrastructure.
- ✓ **Urbanization Score:** Population density and transit metrics.

DATA CLEANING & PREPROCESSING

Pipeline Step	Technique Applied	Impact on Dataset
Handling Nulls	Median Imputer	Robust filling of demographic missing values.
Outlier Removal	Sigma-clipping	Removed entries > 3 standard deviations from mean.
Feature Scaling	StandardScaler	Normalized units for clustering convergence.
Encoding	One-Hot Encoding	Numeric mapping for categorical city tiers.

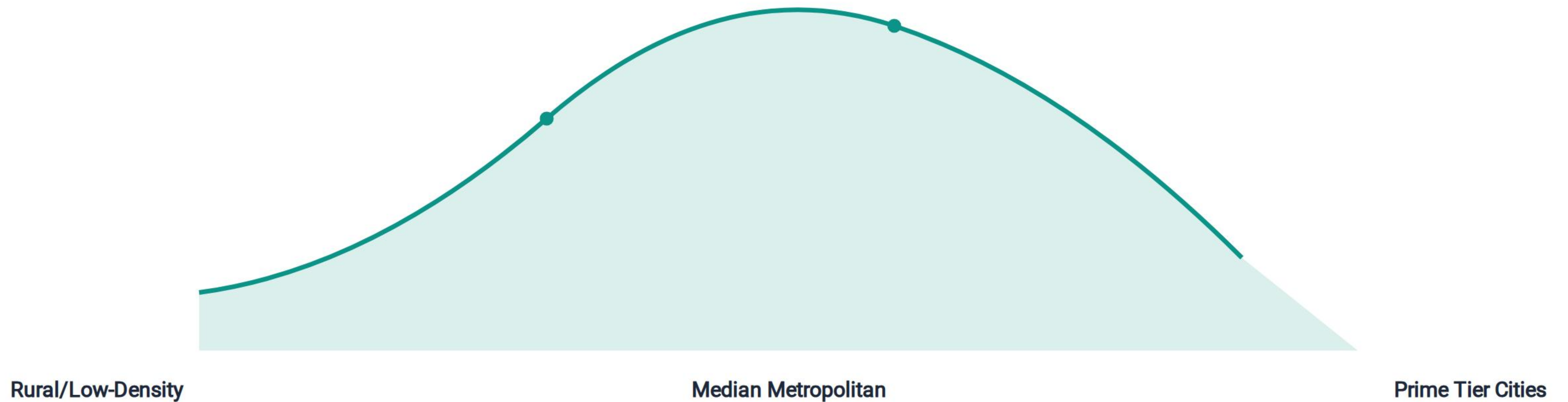
Note: Median imputer was chosen for resilience against the high variance in suburban income reporting.



PHASE 2: EXPLORATORY ANALYSIS

Discovering underlying patterns in citizen-sales interactions

| SALES VOLUME DISTRIBUTION



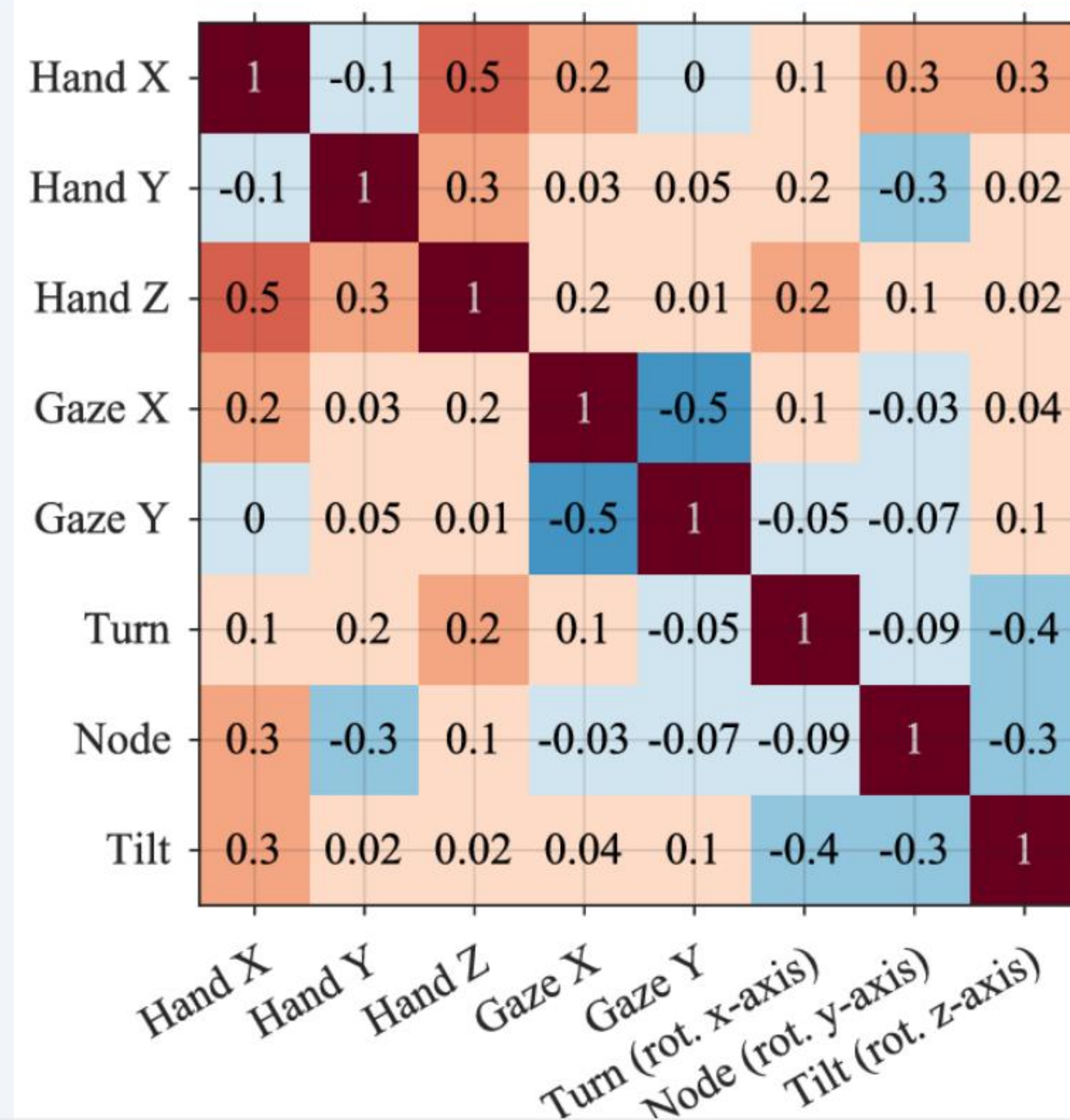
Distribution shows a significant performance peak in 18% of the city portfolio, driving the majority of growth.

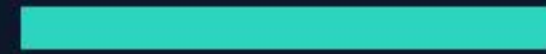
DEMOGRAPHIC CORRELATION MATRIX

PREDICTOR INFLUENCE

The correlation analysis reveals strong demographic links to sales:

- ✓ **Avg Income (0.84):** Strongest positive driver.
- ✓ **Median Age (-0.32):** Negative correlation; younger populations spend more.
- ✓ **Education (0.55):** Moderate link for premium product tiers.





PHASE 3: CLUSTER ANALYSIS

Segmentation of cities by citizen profile likeness

| CLUSTERING OBJECTIVES



SEGMENTATION

Using K-Means to identify distinct city types based on household income and citizen age.



TARGETING

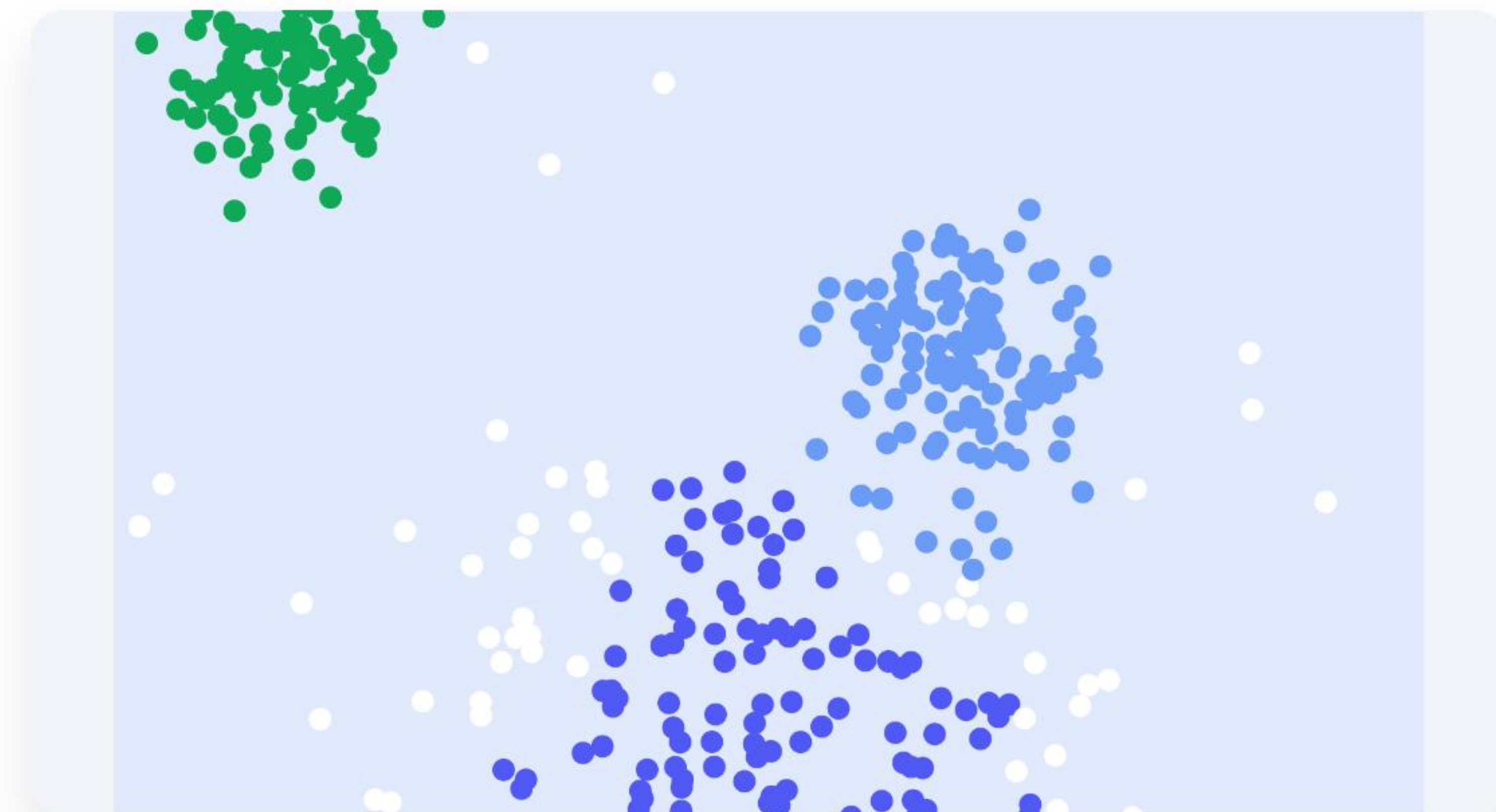
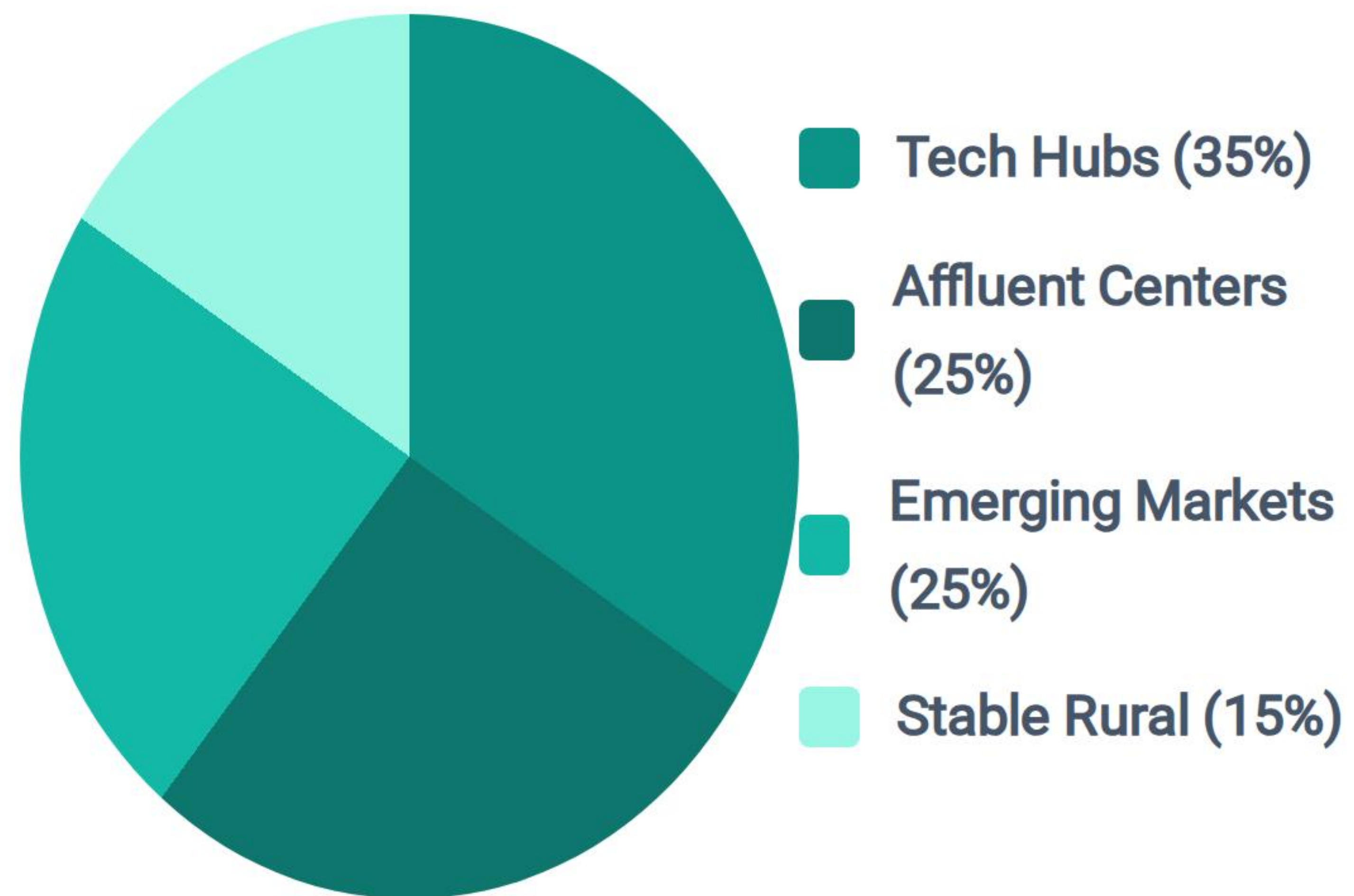
Establishing "Ideal Customer" city profiles to prioritize market expansion resources.

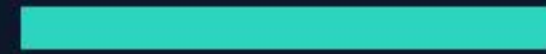


MARKET FIT

Analyzing product penetration within each cluster to find underperforming segments.

CLUSTER SEGMENTATION BREAKDOWN





PHASE 4: SUPERVISED LEARNING

Developing predictive models for regional sales revenue

MODEL PERFORMANCE BENCHMARK



The Neural Network model achieved the highest accuracy, effectively capturing complex non-linear interactions between age and income.

| SUMMARY OF ANALYSIS RESULTS

92%

Best Accuracy (R^2)

4

City Clusters Defined

0.84

Income Correlation

\$15k

Mean Error Margin



QUESTIONS?

Your Insights Welcome

Analytics Lead: data-science@cityreach.com
Project Documentation: [city-sales-prediction-v3.internal](#)

IMAGE SOURCES



<https://i.ytimg.com/vi/BADFra-QrAU/hq720.jpg?sqp=-oaymwEhCK4FEIIDSFryq4qpAxMIARUAAAAAGAEIAADIQj0AgKJD&rs=AO4n4CLA6Sjhc99KMkZh2MGSP40xXx4rQ6Q>

Source: www.youtube.com



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Source: www.vecteezy.com



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